

The National Institute for Forest Products Innovation (NIFPI) was established as an Australian Government initiative in 2018. NIFPI intended to advance research and innovation in the forest and wood products industries addressing issues proposed by industry.

Regional NIFPI centres were established in South Australia, Tasmania and Victoria, with financial support from the Australian Government and the respective state governments, and project specific contributions from industry.

With a strong focus on industry liaison and research adoption, NIFPI aimed to reduce the decline of forestry research and the number of forestry researchers in Australia. The centres aimed to support cutting-edge research, boost innovation, foster research and development collaboration, and secure greater productivity and profitability of forestry in Australia.

A total of 49 projects were funded across the 3 centres with a combined investment value of \$33.61 million. This includes \$18.45 million of cash and in-kind contributions from industry and research partners.

Despite the success of the program, several projects were delayed by the COVID-19 pandemic and the Australian 2019-20 bushfires. These projects are currently being finalised. Eleven projects are still actively underway and will be delivered in the coming months. Reports for these remaining projects will be published once they are received.

Of the published NIFPI projects, nearly:

- 41% focused on improving resilience, productivity and utilisation of the forest resource.
- 22% focused on utilising data for forest management and tree growing improvements.
- 22% focussed on development of timber and non-timber products and improvements to existing products
- 12% focused on improving forestry worker safety across the value chain.

NIFPI outcomes

The program has helped to:

- accelerate forest and wood products research
- strengthen partnerships and collaboration between the Australian Government, state governments, universities and industry
- improve businesses' operations and management of our forests and processing sectors
- provide indirect and spill over benefits in areas such as satellite technology
- develop higher degree and early career researchers in associated research areas
- retain mid-career and senior researchers.

Since 2022-23, the Australian Government has been supporting cutting edge forest industry research and innovation through [Australian Forest and Wood Innovations](#).

The Launceston NIFPI Centre

The Launceston NIFPI Centre research priorities included but were not limited to:

Solutions for the optimal use of dense, remotely acquired data by forest growers

This project, jointly funded with the Mount Gambier NIFPI centre used remote sensing data for better decision making in resource management. It focused on optimising remote sensing technologies for forest resource assessment, health monitoring, and data management. This research involved a team of scientists, tech specialists, and forestry workers working together to enable the application of this technology in real-world forestry operations.

Conceptualise and develop a functioning model for collaborative integrated pest management within the Tasmanian forest industry

This project aimed to develop an integrated pest management model to inform stakeholders on pest and disease status across all land tenures, enabling timely management activities. A Tasmanian Integrated Pest Management Group Steering Committee was established through this project in March 2019.

Increasing the durability, and other material characteristics of Tasmanian hardwoods

This research focused on finding effective ways to improve the durability and fire resistance of Tasmanian hardwoods. It aimed to make these timbers suitable for outdoor use and areas at risk of bushfires. A second goal of the research was to improve Tasmanian hardwoods for indoor use by making them long-lasting, more fire-safe, and less likely to change shape over time.

A forest resource characterisation of Tasmania – Stage 1 of 2. Feasibility

This study assessed the possibility of developing models to characterise the wood volume and wood quality of the Tasmanian hardwood estate. Researchers collected and reviewed available data and information on Tasmanian estates. These models provide valuable information for forest growers, timber producers, and end users.

Eagle Eye – applying the Internet of Things to landscape scale Wedge-tailed eagle management

This project looked at the application of the Internet of Things (IoT) to manage endangered Tasmanian wedge-tailed eagle in a landscape shared with forestry and electricity transmission infrastructure. It aimed to replace costly and risky helicopter checks with remote, real-time data collection. The goal was to help forestry and energy companies make safer, faster, and more wildlife-friendly decisions.

Short log supply chain impacts in hardwood plantations

The study looked at whether adding short saw logs to the usual mix of logs in Tasmanian hardwood plantations would increase or decrease profits. It tested different harvesting methods at 3 sites in northern Tasmania. The goal was to see how short logs affect costs like transport and processing. The study helped to decide if short logs are worth including in timber operations.

Assessing the economic impact of damage to Eucalyptus nitens logs during mechanised harvesting operations

This project assesses improvements to practices and equipment to reduce log damage to Eucalyptus nitens logs during mechanised harvesting. This could improve profitability for landowners, forest growers, contractors and machine manufacturers.

The report is not yet published.

Managing timber's moisture content in the supply chain, construction and in service

This project aimed to understand the moisture content of wood products and timber in the Australian timber supply chain. It found ways to prevent damage and product loss caused by moisture changes. Researchers monitored timber under different conditions and wrapping methods. It developed guidance that allows industry to limit unacceptable moisture content variation and improve customer confidence.

Minimising market-limiting discolouration in appearance Tasmanian hardwood

This project investigated the cause of grey discolouration such as 'tyre track' (a form of blue stain), and 'sticker mark' (light or dark stripes) in Tasmanian blackwood and oak. The goal was to protect the appearance and market value of these timbers. It generated guidance on management and processing options to avoid discolouration.

New methods of reliably demonstrating species durability in commercially relevant timeframes

The focus of this project was to identify faster ways to analyse durability for preservative treated or modified Eucalyptus species timber. It focused on improving testing methods so results could be seen in months or years instead of decades. Researchers trialled lab and field techniques using treated and untreated wood samples. The aim was to build on existing accelerated testing approaches to increase their applicability in the Australian timber industry context.

Implementation of single-step genomic selection in eucalypts

This project used single-step genomic selection with the aim to speed up genetic gain in tree improvement programs for hardwood eucalypts. The project enabled the adoption of genomics

technology into tree breeding and tested its application on an industrial scale for *Eucalyptus globulus* and *Eucalyptus nitens*.

Improve returns to forest owners by exploring the feasibility of a pellet-based industry in Tasmania

This project aimed to understand the practicality of establishing an efficient and profitable wood pelletising industry in Southern Tasmania. It aimed to convert existing wood residues to provide sustainable returns for forest owners.

The report is not yet published.

Sensing technology and digital tools to support decision-making in hardwood timber drying

Few tools currently exist to assist mill and yard managers in their day-to-day drying management decisions. This project aimed to develop and verify a timber drying technology and a decision support application for higher value production across the wider timber industry in Australia and overseas.

Developing a new generation of Tasmanian appearance hardwood products for in-state design and manufacturing

Eucalyptus Nitens is rarely used as a high-end architectural product due to some concerns. These are mainly its low density, and weaker performance compared to other hardwoods. This project aimed to create new engineered wood products using Tasmanian plantation-grown *Eucalyptus nitens*.

Developing laminated structural elements from fibre-managed plantation hardwood

This project aimed to develop reliable methods and expertise for making glue-laminated timber (GLT) from plantation-grown hardwood boards. It focused on grading, jointing, and gluing boards from *Eucalyptus nitens* to create strong structural timber. The goal was to turn lower-value plantation wood into high-value engineered timber products.

Unlocking financial innovation in forest products with natural capital

This project explored how forestry businesses can better manage and benefit from natural capital, like water, soil, and biodiversity. The project aimed to support investment in the non-timber natural capital value of Australian forest resources, by developing methods and tools for forest owners and investors to assess, monitor and manage natural capital risks and opportunities.

Optimising machinery configurations for profitable harvesting operations of small-scale plantations

This project aims to create easy-to-follow guidelines and an online decision support tool to effectively select harvesting equipment configurations for smaller, more spread-out plantation area. This may

improve profitability for landowners, contractors, consultants, forest companies and potential machinery co-operatives.

The report is not yet published.

The Gippsland NIFPI Centre

The Gippsland NIFPI centre research priorities included but were not limited to:

Improving the Commercialisation of Post and Plate Mass Timber Building Systems via Digital Twin Simulation

This project aimed to improve construction efficiency and cost-effectiveness using two advanced mass timber systems—RothoBlaas' Spider and Pillar system, and XLam Australia's Band Beam. Through digital modelling and consultation with experienced builders, the team refined installation methods to reduce on-site work and avoid unreliable adhesive joins. These improvements support safer, faster construction and highlight the potential of timber systems as alternatives to traditional concrete structures.

Autonomous detection and deterrent devices for browsing management in forested landscapes

This project aimed to develop and test automatic surveillance, detection and deterrent mechanisms to assess deer use of heterogeneous forested landscapes and develop new options to reduce the impacts caused by deer.

The report is not yet published.

Harnessing the power of airborne high spatial resolution hyperspectral imagery for the softwood plantation industry

Using advanced sensors on a small aircraft, this project will develop a method to spot nutrient shortages and stress in radiata pine plantations.

The report is not yet published.

Understanding Soil Resources for Radiata Pine Plantation productivity

This project combines Australian soils and climate data with soils and growth data from Gippsland. The datasets will enable growers to use freely available soils data, saving costs associated with field-based data collection.

The report is not yet published.

A feasibility study of a commercially available remote-controlled and high-rate tree pruning unit to create fuel modified zones in softwood plantations

This project evaluates the use of remote-controlled high pruning technology in softwood plantations to create zones with reduced fuel, lowering fire risk and improving fire suppression. It will also examine the occupational health and safety aspects, and the costs involved.

The report is not yet published.

Sustained productivity gains in softwood plantations through enablement of single-step genomic selection

This project was part of a series of research projects that aimed to double the rate of genetic improvement in pine breeding in Australia. It focussed on the Gippsland area, using advanced genomic selection methods to improve pine breeding and ensure the accuracy of pine pedigrees and identities.

Investigation of preservative treated plantation timber fencing and sleeper markets in bushfire prone areas

This project aimed to ensure the safe use of treated timber for fencing and sleepers in areas prone to bushfires. Researchers developed technical guidelines to confirm the effectiveness of these materials. The project recommended using smart design instead of banning timber and called for more research on fire-safe treatments.

Fire, acoustic and structural performance of prefabricated plantation Shining Gum advanced timber composite (ATC) floor systems

The project aimed to develop an innovative floor system using Gippsland's plantation shining gum. Using glulam technology and Computer Numerical Control machines, strong and precise timber parts were created. The system passed tests for strength, fire safety, and sound control thereby meeting building code standards. This provides opportunities for industry to use available resource in innovative timber building systems.

Evaluation of commercial opportunities for laminated veneer lumber from underutilized small diameter resource in Gippsland

This project focuses on testing the potential to process smaller logs from various sources in Gippsland. It uses spindleless lathe technology to create laminated veneer lumber (LVL) and explore its use in residential and commercial construction.

The report is not yet published.

Exploring processing methodologies for production of laminated veneer lumber using under-utilised forest resources in Victoria

This project explored methodologies to heat logs, feedstock testing protocols and make more LVL products using spindleless lathes. This will support the use of small logs to make LVL products.

The report is not yet published.

Strengthening forestry and bioeconomy connections through efficient energy production and use

This project assesses the feasibility of scalable energy systems that use biomass by-products from wood processing mills.

The report is not yet published.

Innovative nursery management solutions to sustainably manage root disease, improve nursery utilization, and enhance resilience and productivity of planted pines

This project aimed to create strategies to prevent root rot disease in nursery stock and stop its spread to plantations. This project helped improve health and survival of softwood seedlings in nurseries. It tested better ways to use fertilisers, fungicides, and microbes. It identified the right balance of fertiliser, fungicide, and beneficial microbes to keep stock healthy.

Identify, measure and model critical elements required of plantation forestry to maintain a social license for plantation operations and expansion in Gippsland

This project assessed the social acceptance of expanding pine plantations in Gippsland. As industry shifts from native forest harvesting and transitions to plantation timber in the Gippsland region, gaining community approval is crucial. The findings will guide future forest industry practices and help maintain community support.

The Mount Gambier NIFPI Centre

The Mount Gambier NIFPI Centre research priorities included but were not limited to:

Research, development and validation of 8-star rated architectural products maximising the use of out-of-grade timber

This project aims to design and create various architectural products that use lower-grade timber. These products will improve manufacturing abilities in architecture, especially to make parts that meet high energy efficiency standards.

The report is not yet published.

Development of a timber industry sector framework for setting carbon emissions targets using the Science Based Targets initiative

The main goal of this project was to create a science-based method to help Australia's wood product industry set clear and achievable carbon reduction targets. The project created targets to reduce greenhouse gas emissions throughout the timber's life cycle. These targets follow international best practices and support timber to maintain its status as a preferred product.

Wearable sensors for improving occupational health and safety of workers in the forestry industry: a pilot prototype for harvesting and processing operations

This project looked at using smart sensors in workwear to improve safety for forestry workers. These sensors can provide early warnings about dangers to workers. An image-based detection system designed as a mobile app was created that could be mounted on safety gear like helmets and vests. The system worked well and has the potential to keep forestry workers safer. keep forestry workers safe in the future.

Development of best practice guidance for protective guarding of mobile plant used in Australian forests

This project aimed to create a clear safety guide for protecting cabins in forestry machines, based on the best national and international standards. The project's industry-based committee gave advice to ensure the safety guide for forestry machines met industry needs. The guidelines developed set out best practice for machine selection, modification, and purchase within Australian forest operations.

Development of best practice fatigue management for the Australian Forest Industry

This project explored fatigue risks and fatigue management system in the forestry sector in Australia. Researchers used surveys, interviews, and data analysis to identify effective coping strategies and guide workplace improvements. The study recommended practical solutions such as flexible scheduling, napping, and fatigue-monitoring technologies to improve worker wellbeing and safety.

Addressing the safety risks of multiple in-cab "driver assist" devices in the Australian Forestry Industry

This project aims to solve the problem of having multiple driver-assist technologies in log truck cabs, making driving unsafe. The project found ways to improve the safety of operators.

The report is not yet published.

Development of state-of-the art genomic resources for pine breeding to enable single-step genomic selection

This project aimed to modernise pine tree breeding in Australia by using DNA technology to improve tree selection and productivity. Traditionally, breeding focused on increasing fibre production, but

genomics can make this process faster and more accurate. Scientists built a genome map and tested over 900 samples to find useful traits. The project used new methods and genomic resources to select the best trees to plant, making the genetics improvement process faster and cheaper.

Developing more productive plantation trees better adapted to changing environments

This project aimed to improve tree breeding by selecting trees with better growth, wood quality and resistance to diseases. It also prepared the plantation industry for climate change by testing trees in a wider range of environments. Trials were set up across different site types. This helps researchers understand how different tree types perform in various conditions and select the best ones for future planting and breeding.

Enhancing softwood and hardwood plantations site productivity and subsequent operational efficiency by use of an innovative clean-row establishment ‘system’

This project looks at how safe, sustainable, and effective it is to use mechanical treatment to clean up harvest residues by sweeping them into rows.

The report is not yet published.

Future Proofing SA Blue Gum Plantations through improved detection of koalas in early planning and forestry operations

This project evaluates old and new methods and technologies to find koalas in blue gum plantations. The insights from the evaluation will be used to develop and test technologies further.

Development and implementation of forest health and biosecurity systems and protocols based on quantitative pest risk and economic impact assessment

The project develops an integrated forest and biosecurity system in the Green Triangle. This system will use pest spread predictions and data on economic impacts to help control costs, keep plantations profitable, and avoid losing market access.

Scoping an Automated Forest Fire Detection and Suppression Framework

This project studied whether using new automated technology to detect forest fires is helpful and cost-effective. The project created a smart scanner called MOBIVLS to automatically find koalas using infrared cameras. The new method performed better than existing systems and helps protect both koalas and workers.

Evaluation of remote sensing approaches for plantation health surveillance

This project compares satellite and airborne remote sensing methods with traditional forest health assessment methods to understand the relative costs associated with each method. This will help to identify a cost-effective method to monitor plant health.

The report is not yet published.

Weed Identification Using Remote Sensing

This project builds on and value adds to the software - PINT, which detects and locates newly planted seedlings and identifies areas of missing trees. This will help to better manage weeds and reduce herbicide usage.

The report is not yet published.

Optimising the management of plantation, water and environmental assets

This project has been undertaken with the broad aim of improving the understanding and measurement of plantation water use and environmental asset management in the Lower Limestone Coast, South Australia. The project used remote sensing and ground-based monitoring to understand the water use by plantations. The research helped fill important knowledge gaps to improve water planning for forests. The findings will be used in the review of the region's water allocation plan by local authorities.

Options for Operating Efficiently and Sustainably with Forest Water License Rules

This project supported the Lower Limestone Coast forestry industry in adapting to water licensing regulations that may significantly reduce water allocations in key areas. Through collaboration with industry, researchers used financial modelling to assess the impact of water restrictions on hardwood and softwood plantations and explore strategies to adapt to the restrictions.

Plantation water use estimation and measurement for plantation forests

This project improves the tools that estimate how much water plantations use, and study how water stress is related to diseases.

The full report is not published yet.

A development program for future revision of AS/NZS 4063.1:2010 Characterization of structural timber Part 1: Test methods

This project is collecting data to support a future revision of AS/NZS 4063.1:2010 so that the assumptions in AS1720 and AS1684 are valid and less onerous than test methods currently in place.

The report is not yet published.

Characterising Softwood Sawn Products in Australia

The project sought to validate the structural performance of machine-graded softwood to check that it complies with the design standards in AS1720.1-2010. The project confirmed that Australian timber meets or exceeds the design standards.

Source: <https://www.agriculture.gov.au/agriculture-land/forestry/national/national-institute-for-forest-products-innovation>